

Question

Write a Java program to demonstrate the relationship **I1 extends I2**. Use constructors, method overriding and method overloading to calculate the area of a triangle.

Steps

1. Create interface **Shape**.
2. Declare a display method.
3. Create interface **Triangle**.
4. Make **Triangle** extend **Shape**.
5. Declare a calculation method in **Triangle**.
6. Create class **TriangleArea**.
7. Make **TriangleArea** implement **Triangle**.
8. Declare base and height.
9. Create a constructor to initialize them.
10. Implement the inherited interface methods.
11. Calculate the area of the triangle.
12. Overload the calculation method.
13. Create an object in **main()**.
14. Call the methods.
15. Display the result.

Algorithm

START

1. Create interface **Shape**.
2. Declare **display()** method.
3. Create interface **Triangle** extending **Shape**.
4. Declare **calculate()** method in **Triangle**.

5. Create class TriangleArea implementing Triangle.
6. Declare Base B and Height H.
7. Initialize B and H using constructor.
8. Implement display().
9. Implement calculate().
10. Calculate Area = $1/2 \times B \times H$.
11. Overload calculate() with different parameters.
12. Create TriangleArea object in main().
13. Call the methods.
14. Display the area.

STOP

```
ExceptionMessages' '-cp' 'C:\Users\venik\AppData\Roaming\Code\User\workspace  
i1extendsI2_68bf330b\bin' 'I1extendsI2'  
Base: 10  
Height: 6  
Area: 30.0  
Area using overloaded method: 30.0  
PS C:\Users\venik\OneDrive\Assignmets\Assignment2\i1extendsI2>
```